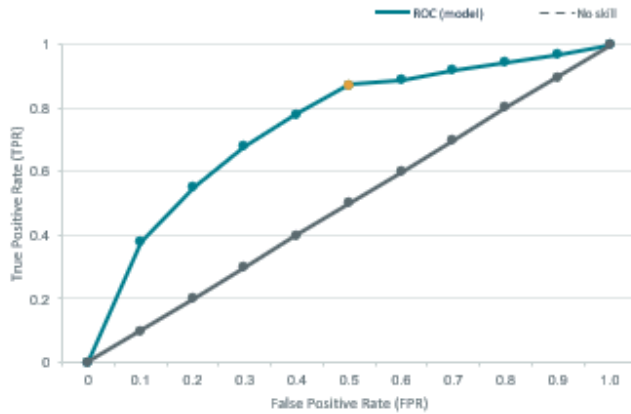


# ROC Curve & AUC

How well a classifier separates the two classes across every possible threshold



**0.75**

AUROC — fair separation

At threshold  $t = 0.50$ : true-positive rate is 87.5% and false-positive rate is 50%.

An AUROC of 0.75 indicates useful, above-random class separation.

## How It's Plotted

X-axis — False Positive Rate:

$$FPR = FP / (FP + TN)$$

Y-axis — True Positive Rate (Recall):

$$TPR = TP / (TP + FN)$$

## Reading the Curve

- Closer to the top-left corner → better performance
- Along the diagonal → random guessing, no discrimination
- Farther above the diagonal → stronger model

## AUC Interpretation

- **1.00** Perfect classifier
- **0.90–0.99** Excellent
- **0.80–0.89** Good
- **0.70–0.79** Fair
- **0.60–0.69** Poor
- **0.50** Random guessing
- **< 0.50** Worse than random

# The Confusion Matrix

Evaluating a classifier by comparing predicted vs. actual labels

| Actual \ Predicted | Predicted Positive                                              | Predicted Negative                                      |
|--------------------|-----------------------------------------------------------------|---------------------------------------------------------|
|                    | TP<br>True Positive<br>Correctly predicted positive             | FN<br>False Negative<br>Missed positive (Type II error) |
| Actual Positive    |                                                                 |                                                         |
| Actual Negative    | FP<br>False Positive<br>Wrongly flagged positive (Type I error) | TN<br>True Negative<br>Correctly predicted negative     |

## Why it matters

- More informative than accuracy alone on imbalanced datasets
- Shows exactly which errors a model makes
- Guides the right metric to prioritize — e.g. recall for disease detection, precision for spam filtering

## Accuracy

$$(TP + TN) / (TP + TN + FP + FN)$$

Overall proportion of correct predictions

## Precision

$$TP / (TP + FP)$$

Of predicted positives, how many are truly positive

## Recall (Sensitivity)

$$TP / (TP + FN)$$

Of actual positives, how many were correctly identified

## Specificity

$$TN / (TN + FP)$$

Of actual negatives, how many were correctly identified

## F1-Score

$$2 \times (Precision \times Recall) / (Precision + Recall)$$

Harmonic balance of precision and recall

# Possible Short Questions (Confusion Matrix)

## Q1. What is a Confusion Matrix?

বাংলা

Confusion Matrix হলো একটি Table যেখানে Actual এবং Predicted Result তুলনা করে Model-এর Performance Evaluate করা হয়।

### English (Exam)

A confusion matrix is a table used to evaluate a classification model by comparing actual and predicted labels.

---

## Q2. What does TP stand for?

বাংলা

True Positive

### English

True Positive

---

## Q3. Define True Positive.

বাংলা

বাস্তবে Positive ছিল এবং Model-ও Positive Predict করেছে।

### English

True Positive is the correctly predicted positive instance.

---

## Q4. What does TN stand for?

বাংলা

True Negative

### English

True Negative

---

## Q5. Define True Negative.

বাংলা

বাস্তবে Negative ছিল এবং Model-ও Negative Predict করেছে।

### English

True Negative is the correctly predicted negative instance.

---

## Q6. What does FP stand for?

বাংলা

False Positive

### English

False Positive

---

## Q7. Define False Positive.

বাংলা

বাস্তবে Negative ছিল কিন্তু Model Positive Predict করেছে।

### English

False Positive is an incorrectly predicted positive instance.

---

## Q8. What does FN stand for?

বাংলা

False Negative

### English

False Negative

---

## Q9. Define False Negative.

বাংলা

বাস্তবে Positive ছিল কিন্তু Model Negative Predict করেছে।

**English**

False Negative is an incorrectly predicted negative instance.

---

## Q10. What is Accuracy?

বাংলা

Model-এর Overall Correct Prediction-এর হার।

**English**

Accuracy is the overall proportion of correct predictions.

---

## Q11. Write the formula of Accuracy.

**English**

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$$

---

## Q12. What is Precision?

বাংলা

Model যাদের Positive বলেছে, তাদের মধ্যে কতজন সত্যিই Positive।

**English**

Precision is the proportion of correctly predicted positive instances among all predicted positive instances.

---

### Q13. Write the formula of Precision.

Precision =  $TP / (TP + FP)$

---

### Q14. What is Recall?

বাংলা

বাস্তবে যত Positive ছিল, Model তাদের মধ্যে কতজনকে ধরেছে।

**English**

Recall is the proportion of actual positive instances correctly identified.

---

### Q15. Another name of Recall?

বাংলা

Recall-এর আরেক নাম Sensitivity।

**English**

Sensitivity.

---

### Q16. Formula of Recall.

Recall =  $TP / (TP + FN)$

---

### Q17. What is Specificity?

বাংলা

বাস্তবে যত Negative ছিল, Model তাদের মধ্যে কতজনকে ঠিকভাবে শনাক্ত করেছে।

**English**

Specificity is the proportion of actual negative instances correctly identified.

---

## Q18. Formula of Specificity.

Specificity =  $TN / (TN + FP)$

---

## Q19. What is F1-score?

বাংলা

Precision এবং Recall-এর Balanced Measure।

**English**

F1-score is the harmonic mean of Precision and Recall.

---

## Q20. Formula of F1-score.

$F1 = 2 \times (Precision \times Recall) / (Precision + Recall)$

---

## Q21. Why is Confusion Matrix important?

বাংলা

কারণ এটি শুধু Accuracy নয়, Model কী ধরনের ভুল করছে তাও দেখায়।

**English**

It shows the types of prediction errors and provides more information than accuracy alone.

---

## Q22. Why is Confusion Matrix preferred over Accuracy?

বাংলা

Imbalanced Dataset-এ Accuracy বিভ্রান্তিকর হতে পারে।

**English**

It is more informative than accuracy, especially for imbalanced datasets.

---

### **Q23. Which metric is important for disease detection?**

বাংলা

Recall

**English**

Recall (Sensitivity).

---

### **Q24. Which metric is important for spam filtering?**

বাংলা

Precision

**English**

Precision.

---

### **Q25. Which error is Type I Error?**

বাংলা

False Positive

**English**

False Positive.

---

### **Q26. Which error is Type II Error?**

বাংলা

False Negative

**English**

False Negative.

---

# Mini Math (Exam Style)

## Math 1

Given:

$$TP = 40$$

$$TN = 50$$

$$FP = 5$$

$$FN = 5$$

Find Accuracy.

### Solution

Accuracy

$$=(40+50)/(40+50+5+5)$$

$$=90/100$$

$$=90\%$$

---

## Math 2

Given

$$TP=80$$

$$FP=20$$

Find Precision.

### Solution

Precision

$$=80/(80+20)$$

$$=80\%$$



---

## Math 3

Given

TP=90

FN=10

Find Recall.

### Solution

Recall

$$=90/(90+10)$$

$$=90\%$$

---

## Math 4

Given

TN=95

FP=5

Find Specificity.

### Solution

Specificity

$$=95/(95+5)$$

$$=95\%$$

---

## Math 5

Given

Precision = 80%

Recall = 80%

Find F1-score.

### Solution

F1

$$= 2 \times 0.8 \times 0.8 / (0.8 + 0.8)$$

$$= 0.8$$

$$= 80\%$$

---

★ আমার **Prediction** (এই **Slide** থেকে সবচেয়ে বেশি আসতে পারে)

#### ● Very Important

1. Define Confusion Matrix.
2. Define TP.
3. Define FP.
4. Define FN.
5. Define TN.
6. Formula of Accuracy.
7. Formula of Precision.
8. Formula of Recall.
9. Formula of Specificity.
10. Formula of F1-score.
11. Why is Confusion Matrix important?
12. Why is Accuracy not enough for imbalanced datasets?
13. Which metric is important for disease detection?
14. Which metric is important for spam filtering?
15. Type I Error.
16. Type II Error.

চমৎকার। এখন আমরা **Slide 2 (ROC Curve & AUC)** থেকে Exam-Oriented Short Questions করব।  
আমি শুধু তোমার **Slide**-এ যা আছে সেটার ওপর ভিত্তি করেই Question তৈরি করছি।

---

## Slide 2: ROC Curve & AUC

### Q1. What is ROC Curve?

বাংলা

ROC Curve হলো এমন একটি Graph যা বিভিন্ন Threshold-এ Classifier দুইটি Class-কে কত ভালোভাবে আলাদা করতে পারে তা দেখায়।

### English (Exam)

ROC (Receiver Operating Characteristic) Curve shows how well a classifier separates two classes across different thresholds.

---

## Q2. What does ROC stand for?

বাংলা

Receiver Operating Characteristic

### English

Receiver Operating Characteristic

---

## Q3. What is AUC?

বাংলা

AUC হলো ROC Curve-এর নিচের Area।

### English

AUC (Area Under the Curve) is the area under the ROC curve.

---

## Q4. What does AUC measure?

বাংলা

Model দুইটি Class-কে কত ভালোভাবে আলাদা করতে পারে তা Measure করে।

### English

AUC measures the classifier's ability to distinguish between two classes.

---

## Q5. What is the X-axis of the ROC Curve?

বাংলা

False Positive Rate (FPR)

**English**

False Positive Rate (FPR)

---

## Q6. What is the Y-axis of the ROC Curve?

বাংলা

True Positive Rate (TPR) অথবা Recall

**English**

True Positive Rate (TPR) or Recall

---

## Q7. Write the formula of False Positive Rate (FPR).

**English**

$$FPR = FP / (FP + TN)$$

বাংলা

Healthy/Negative Sample-এর মধ্যে কতগুলোকে ভুল করে Positive বলা হয়েছে।

---

## Q8. Write the formula of True Positive Rate (TPR).

**English**

$$TPR = TP / (TP + FN)$$

বাংলা

## Q9. Another name of TPR?

বাংলা

Sensitivity বা Recall

**English**

Recall (Sensitivity)

---

## Q10. What does a ROC curve near the top-left corner indicate?

বাংলা

Model-এর Performance খুব ভালো।

**English**

A ROC curve closer to the top-left corner indicates better classification performance.

---

## Q11. What does the diagonal line in ROC represent?

বাংলা

Random Guessing

**English**

The diagonal line represents random guessing (no discrimination).

---

## Q12. What does "No Skill" mean?

বাংলা

Model Random Guess করছে।

### English

The model performs no better than random guessing.

---

### Q13. What does an AUROC of 0.75 indicate?

বাংলা

Model Fair বা Useful Class Separation করতে পারে।

### English

An AUROC of 0.75 indicates fair and useful class separation.

**Note:** তোমার Slide-এ 0.75-এর Example "fair separation" হিসেবে দেখানো হয়েছে।

---

### Q14. What is the AUROC of a perfect classifier?

বাংলা

1.00

### English

1.00

---

### Q15. What AUROC value indicates random guessing?

বাংলা

0.50

### English

0.50

---

### Q16. Which AUROC range indicates Excellent performance?

বাংলা

0.90–0.99

**English**

0.90–0.99

---

**Q17. Which AUROC range indicates Good performance?**

বাংলা

0.80–0.89

**English**

0.80–0.89

---

**Q18. Which AUROC range indicates Fair performance?**

বাংলা

0.70–0.79

**English**

0.70–0.79

---

**Q19. Which AUROC range indicates Poor performance?**

বাংলা

0.60–0.69

**English**

0.60–0.69

---

## Q20. What does AUROC less than 0.50 indicate?

বাংলা

Random Guess-এর চেয়েও খারাপ।

**English**

Worse than random.

---

## Mini Math (ROC & AUC)

### Math 1

Given:

$$FP = 20$$

$$TN = 80$$

Find FPR.

### Solution

$$FPR = FP / (FP + TN)$$

$$= 20 / (20 + 80)$$

$$= 20 / 100$$

$$= 0.20$$

**Answer = 0.20 (20%)**

---

### Math 2



Given:

$$TP = 90$$

$$FN = 10$$

Find TPR.

### **Solution**

$$TPR = TP/(TP+FN)$$

$$=90/(90+10)$$

$$=90/100$$

$$=0.90$$

**Answer = 0.90 (90%)**

---

## **Math 3**

A model has an AUROC of **0.95**.

**Question:** How would you interpret it?

বাংলা

Excellent Performance।

### **English**

The model has excellent classification performance.

---

## **Math 4**

A model has an AUROC of **0.50**.

**Question:** Interpret the result.

বাংলা

Model Random Guess করছে।

**English**

The model performs like random guessing.

---

## ★ এই Slide থেকে সবচেয়ে সম্ভাব্য Exam Questions

### ● Very Important

1. Define ROC Curve.
  2. What does ROC stand for?
  3. Define AUC.
  4. Formula of FPR.
  5. Formula of TPR.
  6. X-axis of ROC Curve.
  7. Y-axis of ROC Curve.
  8. What does the top-left corner indicate?
  9. What does the diagonal line indicate?
  10. AUROC = 1.00 means?
  11. AUROC = 0.50 means?
  12. AUROC ranges (Excellent, Good, Fair, Poor).
  13. Small calculation on FPR or TPR.
- 

## 📌 আমার Overall Prediction

এই দুইটি Slide (Confusion Matrix + ROC & AUC) মিলিয়ে যদি Short Quiz হয়, তাহলে সবচেয়ে বেশি সম্ভাবনা আছে:

- Definitions (Confusion Matrix, Precision, Recall, ROC, AUC)
- Formula (Accuracy, Precision, Recall, Specificity, FPR, TPR)
- TP/TN/FP/FN চেনা
- AUROC Interpretation
- ১-২টি ছোট Calculation

এই Topic দুটো ভালোভাবে পারলে এই অংশ থেকে আসা Short Question-এর বড় অংশই Cover হয়ে যাবে।

